

Simulation and Analysis of AlGaIn/GaN HEMTs and Review of SLCFET Architectures

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Abstract—This paper presents a simulation study of Al-GaN/GaN HEMTs and a literature review of Super-Lattice Castellated FETs (SLCFETs). We analyze the DC characteristics of two HEMT structures and perform a parametric study on aluminum concentration, buffer doping, channel length, and gate length.

Index Terms—AlGaIn/GaN HEMT, TCAD simulation, Band diagram, Parametric analysis, SLCFET.

I. INTRODUCTION

Wide bandgap semiconductors, particularly Gallium Nitride (GaN), have revolutionized high-frequency and high-power electronics. The AlGaIn/GaN High Electron Mobility Transistor (HEMT) relies on the formation of a Two-Dimensional Electron Gas (2DEG) at the heterointerface due to spontaneous and piezoelectric polarization, eliminating the need for impurity doping in the channel.

GaN has a bandgap of 3.44 eV [1], and a critical electric field of 3 MV/cm [2]. This makes GaN suitable for high power applications.

II. DEVICE STRUCTURE AND SIMULATION SETUP

In this study, two distinct HEMT architectures were designed and simulated using Silvaco TCAD, in order to observe the effects of device scaling, doping, and layout on the output characteristics.

A. Device A

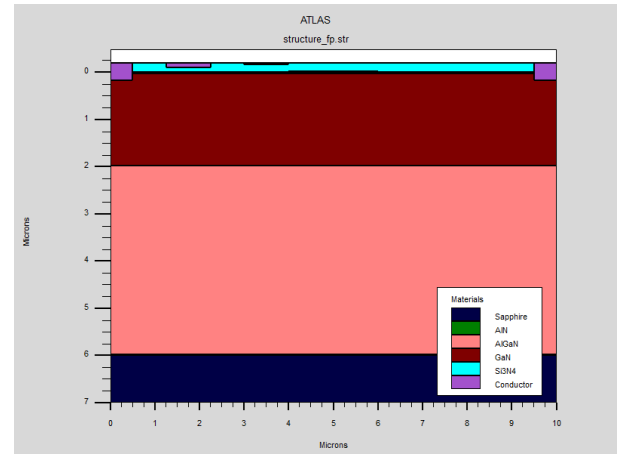
The first device, along with a channel length of 10 μm , features a 0.03 μm $\text{Al}_x\text{Ga}_{1-x}\text{N}$ barrier ($x=0.3$), a 1 μm UID GaN layer, a 4 μm Strain relief layer (SRL) [3], and the substrate used was sapphire. Two source field plates were also added to introduce capacitance between source and gate, in order to reduce the peak field across the junction [4]. The simulation mesh was optimized to capture the high electron density at the hetero interface. None of the regions were doped for this device and no artificial interface charge was given. Rather, models such as polarization and calc.strain were used to simulated the making of the 2DEG. This is also true for the second device.

B. Device B

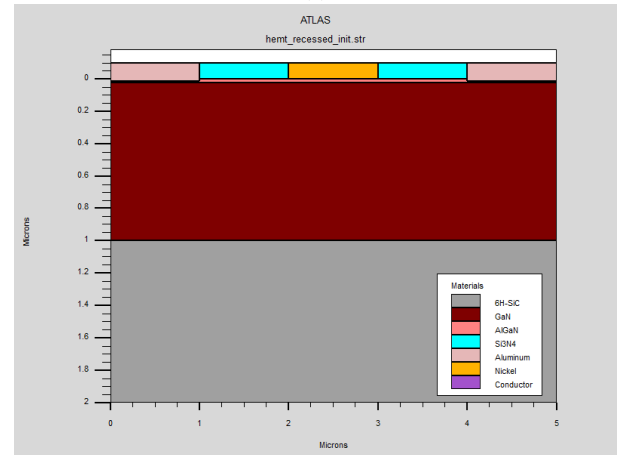
The second device uses a smaller channel length (5 μm), a 0.025 μm $\text{Al}_{0.25}\text{Ga}_{0.75}\text{N}$ barrier, a 1 μm slightly doped GaN layer, and a 6H-SiC layer for the substrate, which offers better

thermal conductivity, for high current applications. A heat sink of sorts was placed below the substrate and lat.temp model was used to define the thermal conductivity of the heat sink which was kept very high to simulate a heat sink. This device was also slightly doped with $10^{15}/\text{cm}^3$ n-type in the GaN layer.

C. Structural Analysis and Band Diagrams



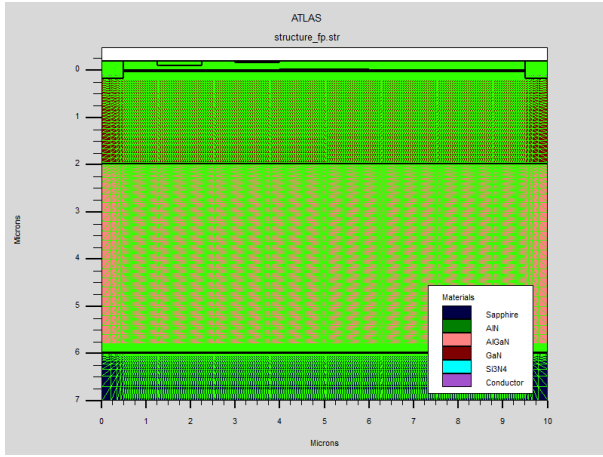
(a)



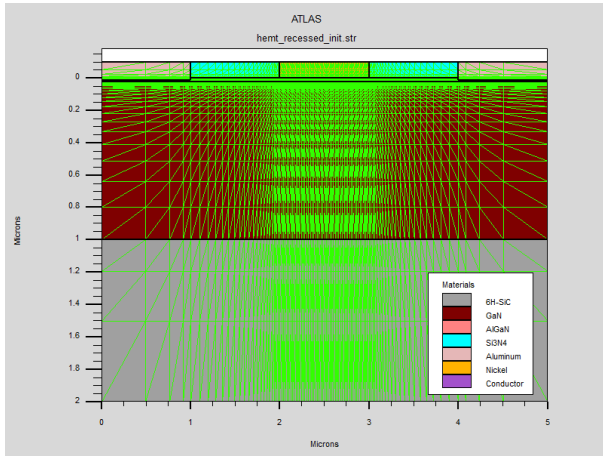
(b)

Fig. 1: Cross-sectional view of (a) Device A and (b) Device B.

Fig. 1 illustrates the physical structure and Fig. 2 the meshing for both devices. The meshing used for A is finer in order to enable placing the source field plates and the SRL.

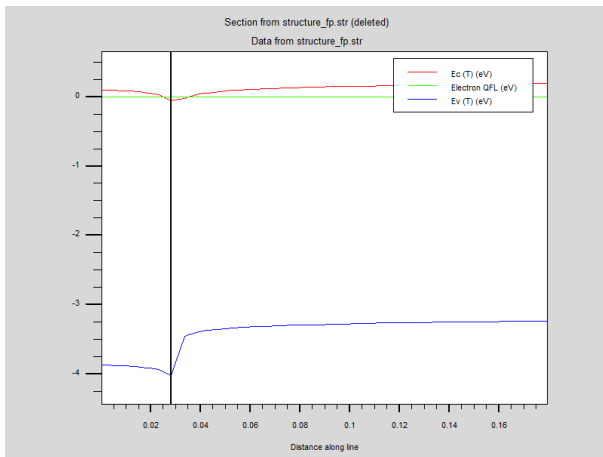


(a)

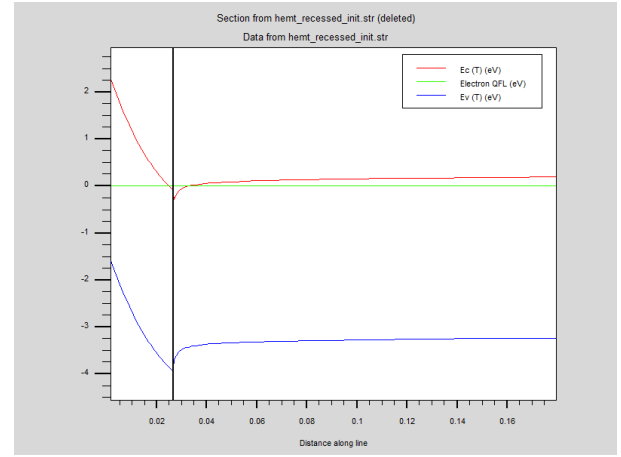


(b)

Fig. 2: Meshing used in (a) Device A and (b) Device B.



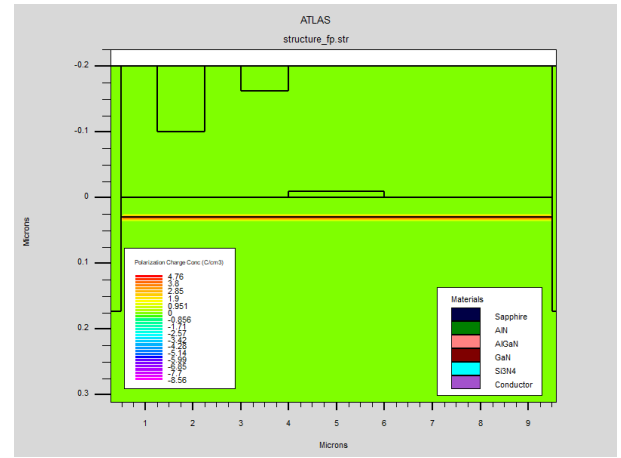
(a)



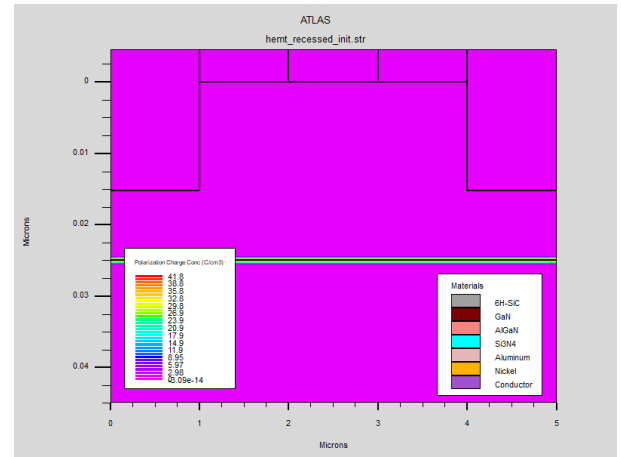
(b)

Fig. 3: Band diagrams obtained for (a) Device A and (b) Device B, showing clearly the formation of a quantum well.

The structural properties obtained after solving for the unbiased equilibrium states are shown in Fig. 3 and Fig. 4.



(a)



(b)

Fig. 4: Interface charges developed in (a) Device A and (b) Device B, due to GaN and AlGaIn's polarization field.

III. DC CHARACTERISTICS AND ANALYSIS

The DC performance of both devices was evaluated through output ($I_D - V_D$) and transfer ($I_D - V_G$) characteristics.

A. $I_D - V_D$ Characteristics

Fig. 5 compares the output characteristics upon varying drain voltage over the range 0-20 volts. Device B exhibits a higher saturation current than Device A, primarily due the following reasons: (a) The channel in B is shorter, which therefore reduces the resistance and increases conductance (read: slope in linear region) offered to the current. (b) The slight doping of the GaN layer in B causes an increase in charge carriers, and therefore causes an increase in current. A stark observation here is the fact that for a brief set of values of V_D in the $V_G = 2$ volts graph in Fig. 5a, the drain current is negative. This is due to the gate contact being a Schottky diode, and will be expanded upon in the next section.

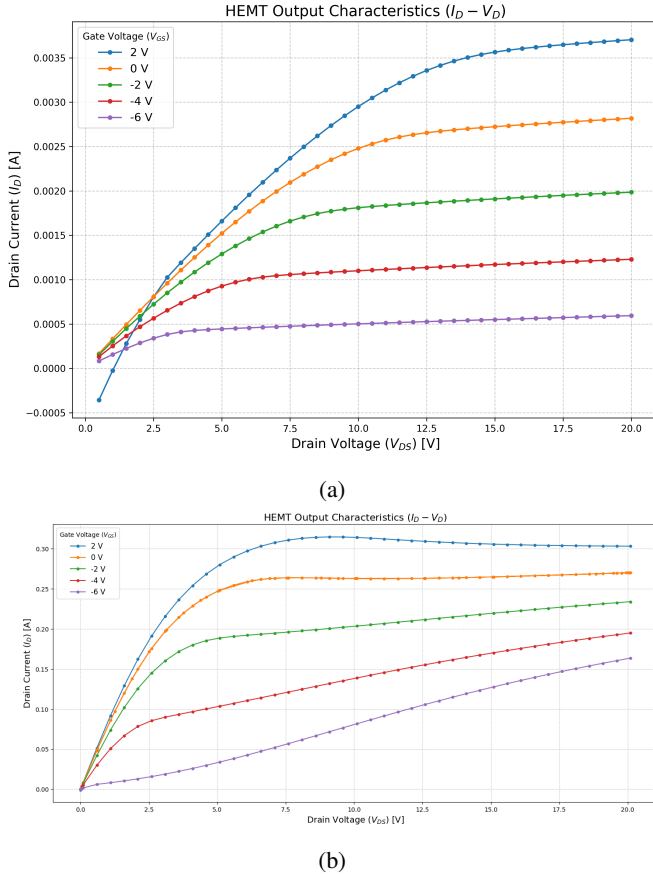


Fig. 5: $I_D - V_D$ characteristics at $V_G = 2, 0, -2, -4, -6$ volts corresponding to (a) Device A and (b) Device B.

B. $I_D - V_G$ Characteristics

The transfer characteristics are presented in Fig. 10. These values were recorded keeping $V_D = 5$ volts. The threshold voltage (V_{th}) for Device A was extracted as ≈ -9 V, and Device B as ≈ -7 V.

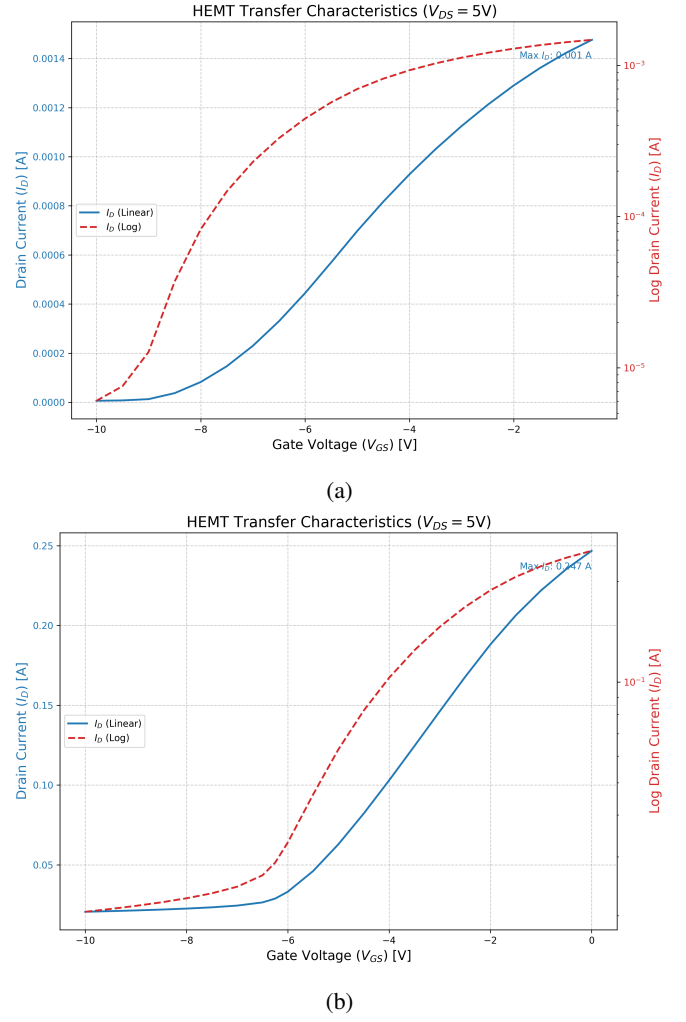
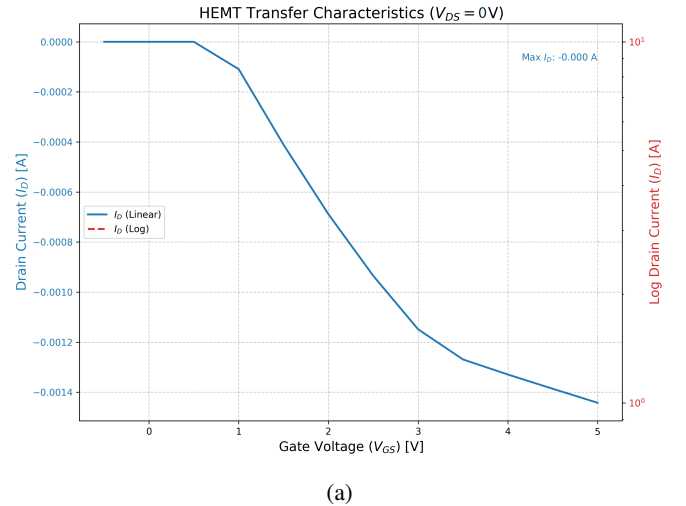


Fig. 6: $I_D - V_G$ characteristics at $V_D = 5$ volts corresponding to (a) Device A and (b) Device B.

If we run a similar sweep keeping V_D at 0 volts, we get the plots as below in Fig. 8.



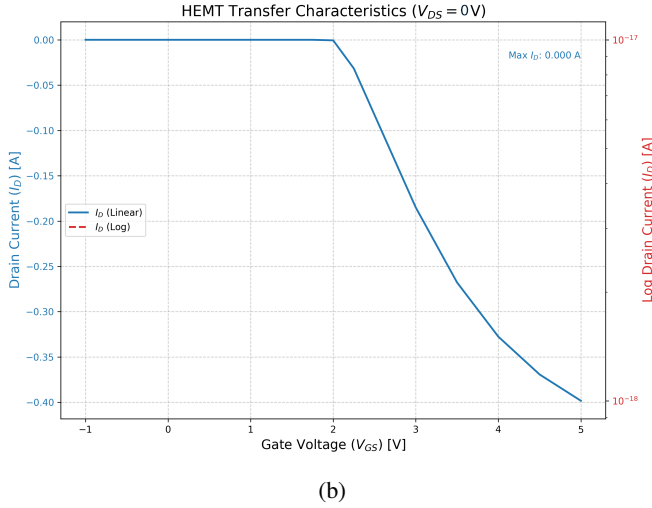


Fig. 8: $I_D - V_G$ characteristics at $V_D = 0$ volts corresponding to (a) Device A and (b) Device B.

It is worth investigating what is happening here. Our hypothesis is that at $V_D = 0$, this current is technically not current from drain to source. This is actually the gate leakage current, which we also observed in Fig. 5a. The Gate-AlGaIn contact is Schottky in nature ($\phi_m > \phi_s$, $\phi_m = 5.1\text{eV}$ (Nickel), $\phi_s = 3.7\text{eV}$) [5]. This is why the gate current is so low normally. However, when the gate voltage is above the Schottky barrier, it starts to conduct an exponentially increasing current. Silvaco somehow treats this as negative drain current, and thus we get such a graph. This stops affecting the drain current at higher drain voltages, and the graph resumes its normal trend.

IV. IMPACT OF ALUMINUM CONCENTRATION

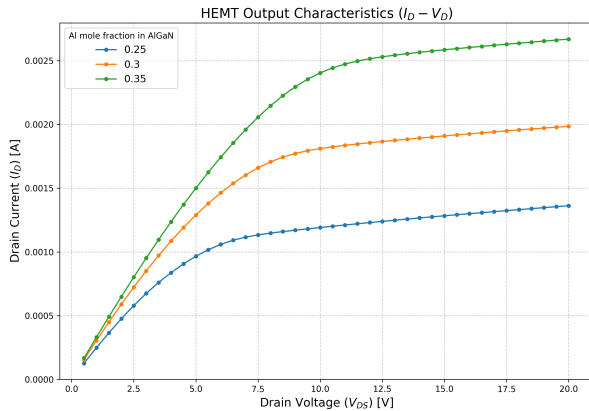


Fig. 9: Effect of Aluminum concentration in the barrier layer on output characteristics

We performed a parametric sweep of the Aluminum mole fraction (x) in the $\text{Al}_x\text{Ga}_{1-x}\text{N}$ barrier of the primary HEMT device (Device A).

As the Al concentration increases, the polarization-induced charge density at the interface increases. The results, shown

in Fig. 9, indicate that a higher Al fraction shifts the threshold voltage to more negative values and increases the maximum drain current.

V. IMPACT OF GAN BUFFER DOPING

The doping concentration of the GaN layer is critical for suppressing leakage current [6]. In this section, the buffer doping (n-type) was varied from 0 to $10^{18}/\text{cm}^3$.

HEMT-like behavior was observed till a doping of $10^{16}/\text{cm}^3$. After this, the conduction was primarily bulk-resistive, following a linear $I_D - V_D$ curve.

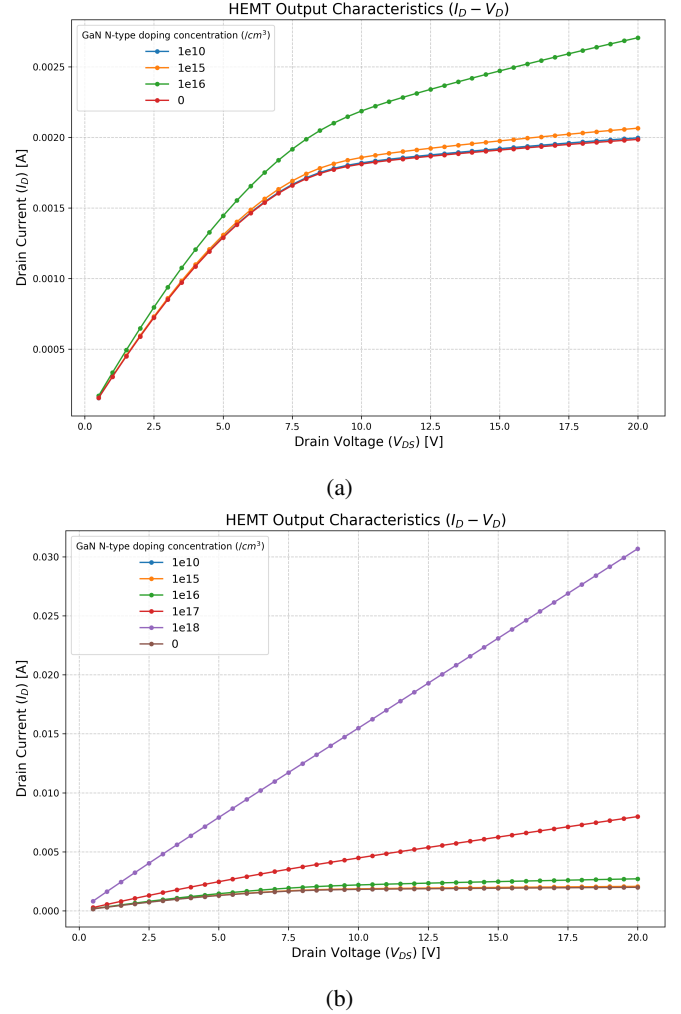


Fig. 10: Output characteristics at different GaN doping levels: (a) Low level, and (b) High level.

VI. IMPACT OF GATE LENGTH SCALING

Gate length (L_g) scaling is essential for high-frequency operation [7]. We simulated the device with varying L_g values.

As L_g decreases, the trans-conductance generally improves. However, Short Channel Effects (SCE) such as Drain-Induced Barrier Lowering (DIBL) become more pronounced at very short gate lengths.

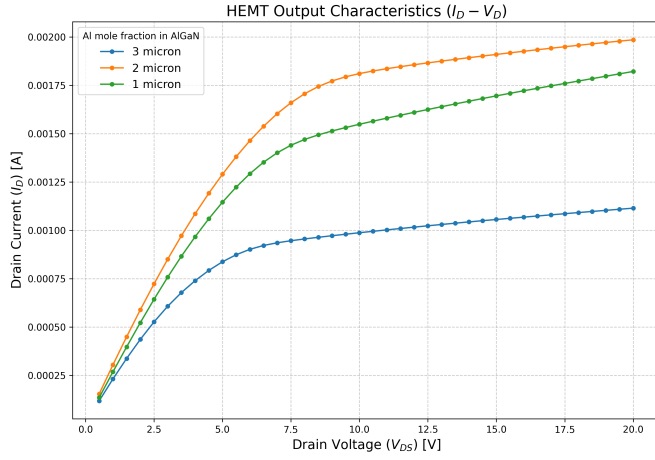


Fig. 11: Gate length (L_g) effects on Output characteristics

VII. REVIEW OF SLCFET TECHNOLOGY

The Super-Lattice Castellated Field Effect Transistor (SLCFET) represents a novel architecture designed to overcome limitations in standard HEMTs, particularly regarding RF switching and linearity. The two main device characteristics which affect the RF applications most are R_{ON} and C_{OFF} which are measured across drain to source during respective states of gate voltage. We need both of these quantities to be as low as possible so that there is minimum loss when the switch is on and there is minimum conduction (especially for higher frequency signals) when the switch is off. The straightforward way to reduce R_{ON} in traditional MOSFETs is to increase the gate width there by increasing the effective area for flow of current. But, this quickly becomes unfeasible as increasing width also increases the area over which the fringing fields between the source and the drain act which increases C_{OFF} . A figure of merit is defined to capture this dependence between R_{ON} and C_{OFF} . Its value is given as-

$$F_{CO} = \frac{1}{2\pi R_{ON} C_{OFF}}$$

Some devices other than MOSFETs which also show promise for this application are the PiN diodes. PiN diodes offer extremely low losses and high isolation, but their power consumption requirements and more complicated control bias networks limit their utility to those applications with relatively few switches [8].

A way to convert the traditional MOSFET design into one with low R_{ON} is by introducing a super-lattice like structures in the body. The super-lattice creates multiple current channels in parallel between the source and drain of the device, lowering ON resistance and the resulting RF switch insertion loss, while maintaining approximately the same OFF capacitance of a conventional HEMT, providing a dramatic improvement in RF switch performance [8]. But these kind of structures face a different problem. The super-lattice structures shields the lower levels from the gate voltage thus preventing their

depletion. This means the channel never completely switches off. In some cases it was seen that increasing gate voltage to high enough values for the channel to close would first result in a breakdown of the FET. To solve this issue of control over the channel the concept of castellations was introduced. Castellations basically act like trenches running perpendicular to the width of the FET. The gate electrode is filled/layered into these trenches. This allows the electric field from the gate electrode to penetrate into the lower layers of the super-lattice thus allowing the gate to control the entirety of the channel. the diagram given in Fig.12 is from [8] and explains this perfectly.

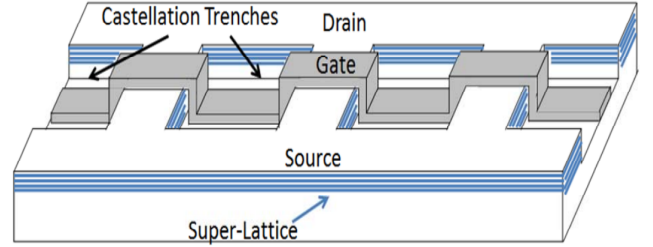


Fig. 12: Schematic of SLCFET architecture.

A. Simulation attempt

We tried to simulate the SLCFET lattice by first starting with just a 2D simulation of a super-lattice. But due to technical errors and the complexity of the simulation we weren't able to move forward with it. The following figure shows the structure we could get for the super lattice.

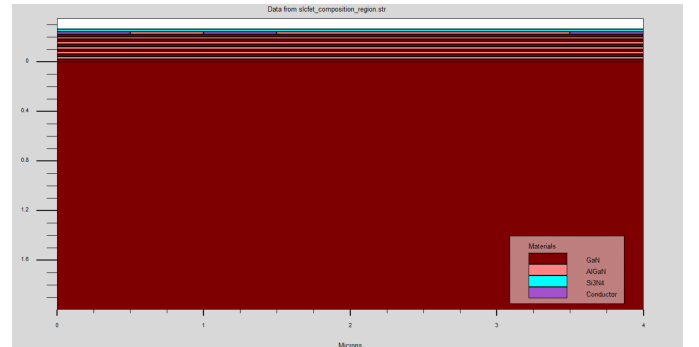


Fig. 13: Super-lattice attempt structure.

VIII. CONCLUSION

This work successfully characterized two AlGaIn/GaN HEMT designs, highlighting the trade-offs in device geometry. The parametric analysis demonstrated the critical role of Al concentration and geometric scaling in tuning device performance. Finally, the review of SLCFETs highlights their potential as a superior alternative for future high-performance RF switching applications.

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